QoE-Aware DRL for Slice Admission Control: A Live Open RAN Testbed Validation

Manoj Prasad Kunasegran
School of Engineering
Taylor's University
Subang Jaya, Malaysia
orcid.org/0009-0005-3169-1873

Wai Leong Pang*
School of Engineering
Taylor's University
Subang Jaya, Malaysia
orcid.org/0000-0001-8407-5648
Corresponding author

Swee King Phang
School of Engineering
Taylor's University
Subang Jaya, Malaysia
orcid.org/0000-0002-7877-8766

***Abstract*—Building on our previous work on Deep Reinforcement Learning (DRL)-driven Slice Admission Control (SAC), which was validated entirely in simulation, this paper moves the same control problem onto a real Open RAN testbed. We deploy a Quality of Experience (QoE)-aware Deep Q-Network (DQN) admission policy on a real OpenAirInterface (OAI) gNodeB (gNB) across three network slices (eMBB, URLLC, mMTC), and compare it against a static baseline and a best-possible static-at-cap policy. After identifying and correcting a calibration issue in our QoE estimation, a fully powered live re-evaluation (46 episodes per arm) shows DQN under the QoE-aware reward sustains 100% SLA compliance throughout, while the SLA-only reward and the static-at-cap arm converge to a statistically indistinguishable rate (44 of 46 episodes fully compliant for each, Fisher exact $p = 1.0$), both modestly above the static baseline's 42 of 46: the collapse-avoidance edge our earlier, smaller-sample evaluation suggested for the SLA-only reward does not hold at this larger, well-powered scale, though the QoE-aware reward's does. We additionally live-evaluate checkpoints trained for a congested, URLLC-preservation scenario: unlike its offline counterpart, where DQN measurably trades eMBB compliance for URLLC under an imposed shared-resource-pool constraint, every arm is fully SLA-compliant live, since this rig's calibrated ceilings are too small relative to the gNB's real capacity for the same forced scarcity to arise – though the QoE-aware reward still yields a materially higher URLLC Mean Opinion Score at the same compliance level. Checking five further independently-trained SLA-reward checkpoints shows live robustness varies substantially by training seed despite equivalent offline convergence, which we trace in part to the offline training environment's own fidelity to real traffic dynamics. We also report the control-loop latency and model footprint needed for practical deployment. These findings demonstrate the feasibility, and surface the honest limits, of deploying DRL-driven SAC on real Open RAN hardware.**

***Index Terms*—5G, Network Slicing, Slice Admission Control, Deep Reinforcement Learning, Quality of Experience, Open RAN**

I. Introduction and Background

Network slicing (NS) lets a single physical 5G deployment serve heterogeneous service classes under per-slice Service Level Agreements (SLAs): Ultra-Reliable Low-Latency Communication (URLLC) requires sub-millisecond latency and high reliability, Enhanced Mobile Broadband (eMBB) demands high throughput, and Massive Machine-Type Communication (mMTC) accommodates large numbers of low-power devices. Slice Admission Control (SAC) decides whether an incoming slice request can be accommodated given current resource availability. Static, threshold-based SAC cannot adapt its allocation as demand shifts, motivating context-aware, learning-based approaches.

Our prior work applied Deep Reinforcement Learning (DRL) to SAC under gNodeB (gNB) congestion, using SLA-driven reward shaping to prioritise URLLC traffic, and later extended this to a third slice (mMTC) with a load-balancing objective across multiple gNBs [1], [2]. Both studies, along with the majority of DRL-driven SAC work surveyed in our systematic review of the 2024–2026 literature [3], were validated entirely in simulation. This paper closes that gap for our own formulation: we realise the same admission-control problem on a real OAI gNB over its native E2 control interface, with the radio channel still simulated (rfsim), and centre the evaluation on Quality of Experience (QoE) rather than SLA compliance alone.

This paper makes the following contributions:

1) We deploy a DRL-based SAC policy on a real O-RAN E2 control loop and identify a structural difference from our earlier simulated work: the real E2 interface exposes no per-flow accept/reject primitive, only a per-slice resource ceiling, so admission must be realised as an adjustment of that ceiling.
2) We compare a QoE-aware reward directly against the SLA-only reward of [1], [2] on the same live hardware.
3) We add a static-at-cap arm that isolates whether the learned policy's advantage is genuine admission timing, or simply the result of using a more generous static allocation.
4) We report the engineering numbers (control-loop latency, model size) a real deployment needs, a calibration issue found and fixed in our QoE estimation, and the fully powered re-evaluation that correction enabled.
5) We evaluate a congested, multi-slice traffic scenario both offline and, at a smaller scale, live, showing where the offline scarcity mechanism does and does not carry over to real hardware.
6) We show that live robustness varies substantially across independently-trained checkpoints despite equivalent of-

fline convergence, and identify a corresponding limitation in the offline training environment itself.

The remainder of this paper is organised as follows: Section II reviews related work on DRL-driven SAC. Section III describes the live testbed, the problem formulation, and the reward function. Section IV presents and discusses the results. Section V concludes the paper and outlines future work.

II. Literature Review

TODO(MANOJ): Literature review to be added here, following the style of Section II in our prior conference papers [1], [2] (narrative summary of ∼10–12 related DRL-driven SAC / real-testbed studies, each introduced with "The authors in [X] proposed..."), concluding with a short paragraph positioning this paper against that related work. Candidate sources are already screened in our systematic review [3].

III. Methodology

A. Testbed

We deploy a real OAI gNB (ORANSlice fork [4]) with a simulated radio channel (rfsim), an Open5GS core, and three `nr-uesoftmodem` user equipment (UE), one per slice. The gNB has a nominal capacity of 106 Physical Resource Blocks (PRBs, $\mu = 1$ numerology). Unlike our earlier simulated work, the real O-RAN E2 interface does not expose a binary accept/reject action; its only control primitive is a per-slice ratio ceiling on a 0–100 scale, enforced by the MAC scheduler [5]. We therefore realise the admission decision as a ceiling adjustment: an accept decision raises a slice's ceiling toward a calibrated maximum, and a reject decision lowers it toward a floor. Ceiling values were calibrated by direct measurement on the live rig rather than assumed, since the 0–100 ratio scale does not map linearly onto real PRBs at this cell's configuration. The static baseline holds its ceiling fixed for the whole episode; a second static arm, *static-at-cap*, instead pins every slice's ceiling at its calibrated maximum from the start, isolating whether a more generous fixed allocation alone can match a learned policy. Learned arms are DQN policies [6], trained offline against a closed-loop simulated traffic source, then evaluated live with frozen weights.

B. Problem Formulation

The gNB state at time t is defined as

$$s_t = \left(\frac{U_s(t)}{B},\ C_t,\ \frac{L_s(t)}{L_{\max}}\ \forall s \in \mathcal{S} \right) \tag{1}$$

where $U_s(t)$ is the resource allocation of slice $s \in \mathcal{S} = \{\text{eMBB}, \text{URLLC}, \text{mMTC}\}$, B is the total resource budget on the ceiling's normalised 0–100 scale, C_t is the current congestion level, and $L_s(t)$ is slice s's queue length, normalised by its maximum capacity $L_{\max}$. The agent's action at each step is the ceiling adjustment described above. The reward function is

$$r(s_t, a_t) = \sum_{s \in \mathcal{S}} \left(\omega_s R_s - \lambda_s \right) - \mu C_t \|a_t\|_1 + \eta\, \mathrm{MOS}_t \tag{2}$$

```
Input: offered_{t−1}, backlog_{t−1}, ceiling_t
mean ← mean_offered_ratio × B × gnb_load
drift ← 0.1 × (mean − offered_{t−1})
offered_t ← max(0, offered_{t−1} + drift + noise)
if a reject freed backlog last step:
   backlog_{t−1} ← max(0, backlog_{t−1} − relief)
demand ← offered_t + backlog_{t−1}
served ← min(demand, ceiling_t)
backlog_t ← min(backlog_cap, demand − served)
loss_t ← 0.02 + 0.3 × (backlog_t / backlog_cap)
emit KPM sample (served, backlog_t, loss_t)
agent observes s_t, sends ceiling_{t+1}
```

Algorithm 1. Offline closed-loop traffic simulation (one slice, one step).

where ω_s and λ_s are slice s's reward weight and SLA-violation penalty, R_s is the reward for serving slice s, and $\mu C_t \|a_t\|_1$ penalises total accepted volume by the current congestion level, matching the SLA-driven reward structure of our earlier work [1], [2]. Setting the QoE weight $\eta = 0$ recovers the SLA-only reward used in that prior work; setting $\eta = 1$ adds a Mean Opinion Score (MOS) term, estimated by a calibrated QoE-scoring model, giving the QoE-aware reward evaluated here. Both variants are compared on the same live hardware. We also report a priority-weighted compliance score, using each slice's declared importance weight rather than an unweighted average across slices of unequal importance.

C. Offline Training Environment

A DQN policy needs on the order of hundreds of episodes of trial and error per seed to converge; across this study's six independently-seeded checkpoints alone that is 1,800 training episodes, far more live rig time than this paper's entire live evaluation uses. Training therefore happens offline, against a synthetic, closed-loop traffic source, before the resulting frozen weights are deployed live with no further learning. The source is closed-loop in the specific sense that matters for learning: served traffic is capped by the slice's own most recently commanded ceiling, and demand the ceiling could not serve persists as backlog into the next step rather than vanishing, so an admission decision carries a real, delayed consequence the agent can learn from rather than being free. This closed-loop property is not incidental: an earlier, open-loop source we also implemented sampled demand and backlog independently every step, recording admission decisions but never feeding them back into future state, under which always accepting was trivially reward-optimal – training against it would validate only the training loop's mechanics (replay buffers, checkpointing, logging), not any genuine admission-demand tradeoff, which is why we do not use it for the results reported here. Algorithm 1 summarises the per-slice, per-step update the closed-loop source runs.

This offline source's demand mean is tied to values measured live on this rig, but its temporal dynamics – the random walk's volatility, how backlog accumulates and drains, and a loss channel derived purely from backlog fraction rather than an independent radio-loss process – are not themselves validated against live traffic. This matters in practice: held-out offline evaluation of the same six checkpoints (100 fresh episodes each, never used in training) does not rank-correlate

TABLE I
LIVE SLA COMPLIANCE (%), 46 EPISODES/ARM

Arm	eMBB	URLLC	mMTC
Static baseline	91.7	91.4	91.4
Static-at-cap	97.4	95.7	95.7
DQN (SLA reward)	96.6	95.7	95.8
DQN (QoE reward)	**100.0**	**100.0**	**100.0**

with their live compliance (Section IV-A) – a checkpoint with a perfect live record can look indistinguishable offline from one that fails live. We traced part of this to a simulator parameter (the backlog's capacity) chosen only to keep trivial baseline policies distinguishable during earlier development, never validated against real SLA-margin magnitude; retargeting it to match measured live margins improved the correlation only marginally and not to statistical significance, so we treat this as a genuine, open limitation rather than a solved calibration bug. Offline training itself still produces competent policies – most checkpoints trained this way perform well live (Section IV-A) – but offline evaluation cannot presently be used to rank or pre-select which checkpoint will, which is why every number in this paper's Results is a live measurement, not an offline projection.

D. Evaluation Setup

Each of the four arms was evaluated live across 11 random seeds, with frozen policy weights: an initial batch of 3 seeds at 2 episodes each, extended in two further rounds of 3 and 5 seeds at the full 5-episode protocol once a QoE-calibration issue discovered during this study (Section IV-B) was corrected, for 46 fully-logged episodes per arm – large enough that a Fisher exact test between two arms tied at their observed rate is a genuine null result, not merely an underpowered one. The static-at-cap arm reuses the static baseline's own control code unmodified, with its ceiling parameter changed only in configuration. Training followed Section III-C's offline procedure, 300 episodes per seed.

IV. RESULTS AND DISCUSSION

A. Live Testbed Results

Table I summarises SLA compliance for all four arms across 46 live episodes each, following a fully powered re-evaluation under the corrected QoE calibration described in Section IV-B. DQN under the QoE-aware reward sustains 100% SLA compliance on every slice, in every episode (46/46 episodes fully compliant). DQN under the SLA-only reward and the static-at-cap arm reach an identical, slightly lower rate (44/46 each); the static baseline trails both at 42/46. Fig. 1 illustrates the underlying mechanism: the baseline's commanded ceiling stays flat at its starting value for the whole episode, while DQN raises each slice's ceiling toward its calibrated maximum within the first few steps and holds it there.

A natural question is whether DQN's advantage simply reduces to using a more generous fixed ceiling. Our earlier, smaller-sample evaluation (15 static-at-cap and a reverified

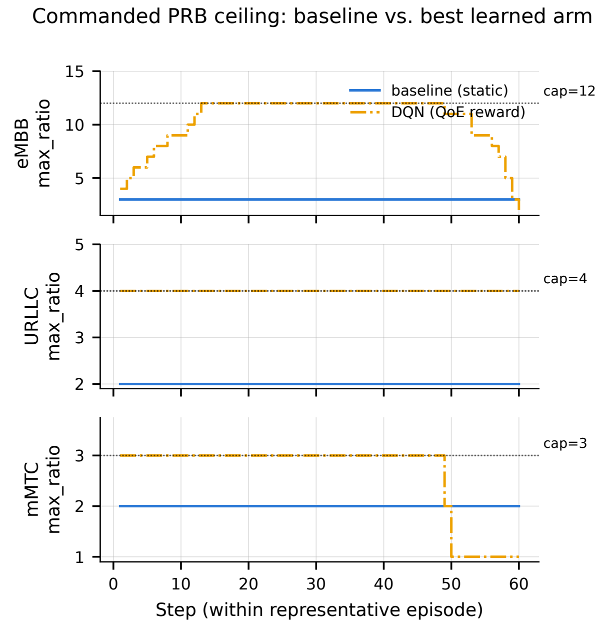

Fig. 1. Commanded ceiling per slice over one representative episode: static baseline (flat) vs. DQN (rises to its calibrated maximum).

25 DQN-SLA episodes) suggested a clear answer: static-at-cap collapsed in 27% of episodes while DQN-SLA never collapsed once (Fisher exact test, $p = 0.0149$). At this larger, fully powered sample, that edge does not hold for the SLA-only reward: DQN-SLA and static-at-cap reach an identical 44/46 fully-compliant episodes (Fisher exact $p = 1.0$) – a large enough sample that this is a genuine null result, not merely insufficient power. Both do, however, clearly separate from the static baseline's 42/46, though not to a statistically significant degree at this sample size (Fisher exact $p = 0.68$ for each vs. baseline). The QoE-aware reward is the one variant whose advantage is unambiguous, with a clean 46/46 record throughout – though even its comparison against the baseline is not itself statistically significant here (Fisher exact $p = 0.12$), since the baseline's own compliance is high enough at this scale to leave less room to separate from. We report both rounds rather than only the more favourable one: a fixed allocation's failure mode is real and shared by every non-learning arm, but the SLA-only reward's learned admission timing does not distinguishably improve on a well-chosen fixed ceiling here, contrary to what our initial, smaller sample suggested; only the QoE-aware reward's advantage is unambiguous, and even that awaits a larger sample to separate cleanly from the baseline.

Fig. 2 shows this episode-by-episode: DQN under the QoE-aware reward sits at exactly 100% in every episode, while the static baseline, the static-at-cap arm, and DQN under the SLA-only reward each show the same bimodal split between fully-compliant and collapsed episodes.

A checkpoint is a single set of frozen DQN weights saved at the end of one offline training run; it is what is actually deployed live, making a state-to-ceiling decision every control cycle with no further learning. Since only

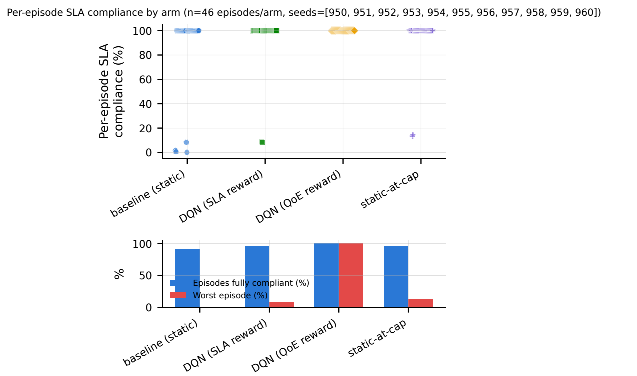

Fig. 2. Per-episode SLA compliance (%). Top: one point per episode. Bottom: fraction of episodes fully SLA-compliant and worst single-episode compliance. Baseline, static-at-cap, and DQN under the SLA-only reward all show the same bimodal split; only DQN under the QoE-aware reward stays uniformly compliant.

one DQN-SLA checkpoint was evaluated above, we trained 5 more independently-seeded DQN-SLA policies under the same procedure and live-evaluated all of them at the same 21-episode protocol. Despite converging equally well offline – indistinguishable Q1→Q4 reward improvement across all six seeds – live records varied substantially (3/5 perfect, one matching the original 19/21, one collapsing to 13/21, worse than baseline): live robustness is not predictable from offline convergence alone, and the checkpoint above is a representative, not best-or-worst-case, draw. This is the practical face of the offline-environment limitation noted in Section III-C: two checkpoints can look equally converged offline and still behave very differently under real traffic.

B. Deployment Feasibility

Measured directly against the real gNB, the control loop's E2 message round trip has a median latency of 0.57 ms, and the DQN model itself (under 20,000 parameters) selects an action in under 70 μs on CPU. Both are a small fraction of the 5-second control interval used in this study, indicating the learned policy adds negligible overhead to the control loop.

Comparing offline training predictions against the live results showed a good match for eMBB traffic, but not for URLLC and mMTC, which were trained under a different demand assumption than what was observed live. During this comparison we also identified a calibration issue in our QoE estimation: the throughput measure used to estimate MOS during training did not match the units produced by the live environment, which understated MOS for well-served traffic. We corrected this calibration and verified it against real observed operating points; Table I above reports the resulting fully powered live re-evaluation.

C. Congested Scenario (Offline)

Sections IV-A and IV-B use one constant demand level per episode. To test a harder, more realistic scenario, we evaluate the same trained policies offline under continuously varying, randomised multi-slice traffic with genuine cross-slice resource contention: when combined demand exceeds the shared PRB pool, every slice's served allocation is scaled down in proportion to its request, rather than each slice being served independently. Pooled over 11 seeds (165 held-out episodes per arm), both DQN variants raise URLLC compliance above the static baseline's 19.7% (to 27.7% for the SLA-only reward, 25.0% for the QoE-aware reward) by actively trading off eMBB (32.7% →7.7%/8.2%) and mMTC (19.8%→9.4%/11.6%) compliance – the same URLLC-first behaviour our earlier simulated work targeted [1], [2], now shown under real cross-slice contention rather than each slice's demand being satisfied independently, and replicating cleanly on an independent 8-seed extension of this evaluation. Weighted by each slice's declared priority, the static baseline still scores higher overall (24.9% vs. 19.1%/17.9% for DQN), since eMBB's importance outweighs URLLC's gain – a real, repeatable tradeoff whose net value depends on whether the priority weights themselves should be revisited, an open question for future work.

We additionally live-evaluated the same congested/URLLC-preservation checkpoints on the real rig, at a smaller pilot scale (one seed, 2 episodes per arm). Unlike the offline result above, every arm – including the static baseline – was fully SLA-compliant live. This is a scale mismatch, not a contradiction: the offline scenario's tradeoff is driven by an imposed shared-resource-pool budget deliberately set smaller than what the three slices' ceilings could jointly request, whereas this rig's calibrated ceilings (summing to a small fraction of the gNB's real 106-PRB capacity) never approach genuine physical contention at these traffic levels, so no forced tradeoff arises live. The QoE-aware reward nonetheless produced a materially higher URLLC Mean Opinion Score (4.83) than the SLA-only reward (2.44) or the static baseline (2.66) at the same 100% compliance – a difference in experience quality the raw compliance numbers alone do not show. A live reproduction of genuine physical multi-slice scarcity would require substantially higher per-slice ceilings and traffic than this campaign's calibration, which we leave for future work.

D. Summary of Findings

At full statistical power, the QoE-aware reward's advantage over static control clearly holds up (46/46 vs. 44/46 for DQN-SLA and static-at-cap, 42/46 for the baseline). The SLA-only reward's own edge, which a smaller-sample round suggested was equally clear, does not replicate; checking 5 more independently-trained checkpoints shows why – live robustness varies by training seed in a way offline convergence does not surface, traced in part to the offline training environment's own limitations (Section III-C). The offline congested scenario still shows URLLC prioritisation consistent with our earlier simulated work, though not yet a net win once priorities are weighted, and does not reproduce live at this campaign's ceiling scale.

V. Conclusion and Future Work

This paper realised the DRL-based slice admission control problem from our prior simulated work [1], [2] on a real OAI gNB over its native E2 control loop, with the radio channel still simulated. After correcting a QoE-calibration issue, a fully powered live re-evaluation (46 episodes/arm) showed the QoE-aware reward sustaining 100% SLA compliance, ahead of the SLA-only reward and static-at-cap (44/46 each – revising our earlier, smaller-sample finding of a significant SLA-only edge, p=0.0149 $\rightarrow$ p=1.0 at full power) and the static baseline (42/46). The control loop adds negligible latency and model overhead. An offline congested-traffic evaluation showed the learned policy can prioritise URLLC as in our earlier simulated work, though not yet as a clear win once slice priorities are weighted; live, at this campaign's ceiling scale, no arm was forced into that tradeoff, though the QoE-aware reward still gave URLLC a materially better experience score.

Our future work will (1) improve the offline training environment's temporal fidelity to real traffic, needed to make offline evaluation predictive of live robustness rather than only a source of competent policies (Section III-C), (2) recalibrate ceilings and traffic to attempt a live reproduction of genuine multi-slice physical scarcity, and (3) extend this single-agent, single-gNB formulation toward the multi-agent, multi-gNB setting proposed in our systematic review [3].

Acknowledgment

This research was supported by the Ministry of Higher Education (MOHE), Malaysia through the Fundamental Research Grant Scheme (FRGS/1/2024/ICT06/TAYLOR/02/1).

References

[1] M. P. Kunasegran, W. L. Pang, and S. K. Phang, "Optimising resource allocation in 5G networks: Balancing URLLC and eMBB traffic under gNB congestion," in *2025 Multimedia University Engineering Conference (MECON)*, 2025, pp. 1–6.

[2] ——, "SLA-aware DRL for joint slice admission control and load balancing in multi-service 5G RAN," in *2025 9th International Conference on Recent Advances and Innovations in Engineering (ICRAIE)*, 2025, pp. 207–212.

[3] ——, "Comprehensive review of optimisation techniques for user-centric distributed network slicing in 5G networks," *IEEE Access*, 2026, vERIFY: final volume/issue/DOI once formally published – title page seen locally still shows placeholder publication-date/DOI fields.

[4] H. Cheng, S. D'Oro, R. Gangula, S. Velumani, D. Villa, L. Bonati, M. Polese, G. Arrobo, C. Maciocco, and T. Melodia, "ORANSlice: An open-source 5G network slicing platform for O-RAN," in *Proceedings of the 30th Annual International Conference on Mobile Computing and Networking (ACM MobiCom)*, 2024, repository: https://github.com/wineslab/ORANSlice. This work's fork: OAI 2024.w28 base, commit b9bcc9b – citation confirmed via the repository's own "how to cite" guidance, 2026-07-30.

[5] O-RAN Alliance, "O-RAN working group 3, near-RT RIC architecture and E2 general aspects and principles (E2AP)," O-RAN Alliance, Tech. Rep., 2023, vERIFY: exact spec number/version (e.g. O-RAN.WG3.E2AP-v03.00) and year.

[6] V. Mnih, K. Kavukcuoglu, D. Silver, A. A. Rusu, J. Veness, M. G. Bellemare, A. Graves, M. Riedmiller, A. K. Fidjeland, G. Ostrovski *et al.*, "Human-level control through deep reinforcement learning," *Nature*, vol. 518, no. 7540, pp. 529–533, 2015.